

3 (a) Define *gravitational potential* at a point.

.....

 [2]

(b) A satellite of mass m orbits in a circular path around the Earth, of mass M and radius R_E , at a height h above the Earth's surface.

(i) On the axes of Fig. 3.1, from $h = R_E$ onwards, sketch and label the variation with h of

1. the gravitational potential energy E_p , [1]

2. the kinetic energy E_k of the satellite. [1]

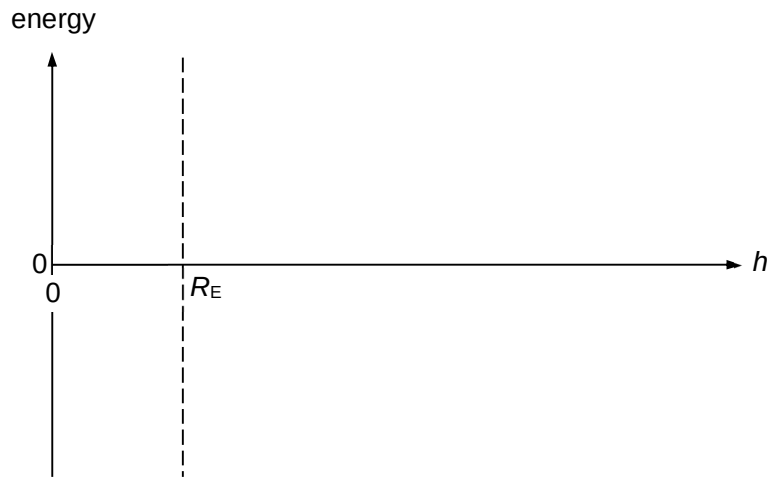


Fig. 3.1

(ii) The Earth has radius 6.4×10^6 m and mass 6.0×10^{24} kg.

A satellite has mass 1200 kg and an orbital period of 12 hours.

Determine h .

$h = \dots\dots\dots$ m [3]

- (iii) State and explain whether work done required on the satellite is positive, negative, or zero, in order for the satellite to orbit at a larger h .

.....

.....

..... [2]

4. (a) Fig. 4.1 shows two coherent sources P and Q producing waves. A wave detector is